



GATE SMASHERS

GATE2024

Engineering Mathematics

Topic: Probability and Statistics

Subtopic:Counting (Restricted Permutation).....

LECTURE - 3

EXPLANATION

RESTRICTED PERMUTATION -

• Permutation of n different things, taken all at a time in which p things always occur together = $p! \cdot (n-p+1)!$



• Permutation of n different things, taken r at a time, when x particular things are always included = ${}^r P_x \times {}^{n-x} P_{r-x}$

• Permutation of n different things, taken r at a time, when x particular things are always excluded = ${}^{n-x} P_r$

Explanation By Examples -

Q1) A student wants to arrange 3 Economics, 2 History and 4 English books on a shelf. If the books on the same subject are different then find,
i) No. of possible arrangements.
ii) No. of possible arrangements when all books on a particular subject are to be together.

Q2) In how many ways can 5 boys and 3 girls be arranged so that no two girls may sit together.

Q3) How many different words can be formed of letter of the word 'COMBINE' so that

i) Vowels always remain together?

ii) all vowels never occur together?

iii) no two vowels are together?

iv) vowels may occupy odd places?